2022 Fall General Chemistry CH105 Midterm Exam Answer

Part A (20 * 2 Pts = 40 Pts)

1-5 ABDCB　　6-10 DBEAA　　11-15 CBDCC　　16-20 CEDBB

Part B (6 * 4 Pts = 24 Pts)

B1)
(a) (2 points)
Molar mass of K_3PO_3 = 39.098 * 3 +30.974 + 4 * 15.998 = 212.26 g/mol
Initial mole of K_3PO_4 = 70.5/1000/212.26 = $3.3\underline{2}14 * 10^{-4}$ mol (3 sig.)
Initial mole of $AgNO_3$ =15.0/1000 * 0.050=$7.\ \underline{5} * 10^{-4}$ mol (2. sig.)
1 mole K_3PO_4 will consume 3 mole $AgNO_3$, $3.3\underline{2} * 10^{-4}$ mole K_3PO_4 will consume $9.9\underline{6} * 10^{-4}$ mole $AgNO_3$. $AgNO_3$ is not enough. Thus, $AgNO_3$ is limiting reactant.
(b) (2 points)
Molar mass of Ag_3PO_4 = 107.87 * 3 + 30.974 + 4 * 15.998 = 418.576 g/mol
Theoretical yield of Ag_3PO_4 = 1/3 * 7.5 * 10^{-4} = $2.5*10^{-4}$ mol (2 sig.)
percent yield of Ag_3PO_4 =89.8/1000/ ($2.5*10^{-4}$* 418.576) * 100% = 85.815% == 86%
B2)
(a) (2 points) increasing radius $Be^{2+} < Mg^{2+} < Ca^{2+}$
(b) (2 points) decreasing acidity $N_2O_5 > CO_2 > SiO_2 > CaO$
B3)
(a) O: 2,　(b) Ge: 2,　(c) Nb: 3,　(d) Mo: 6.　(1 point each)
B4)
(a) $CH_4(g) \rightarrow C(g) + 4\,H(g)$
ΔH_1 = 718.4 + 4 * 217.94 -(-74.8) = 1664.96 kJ = 1665.0 kJ　　(2 pts)
(b) $CH_4(g) \rightarrow C$ (s, graphite) + $2H_2(g)$　　ΔH_2 = 74.8 KJ　　(2 pts)
B5)
(a) Li has larger first ionization energy than Na. Compared with Na, Li has a smaller radius, and its valence electron is more attractive by nucleus.
(b) Magnesium has positive electron affinity. When a electron adds to magnesium atom, it goes to a higher empty p orbital. That's will be energetic unfavorable.
B6)
(a) aluminum hydroxide,　　(b) sulfurous acid,
(c) tetraphosphorus hexasulfide,　　(d) iron(III) carbonate / ferric carbonate

Part C (4 * 9 Pts = 36 Pts)

C1)
(a) (2 points)
$c_{stock}\ V_{stock} = C_{dilute}\ V_{dilute}$　　V_{stock} = 1000.0/1000*0.250/14.8*1000 = 16.8918mL = 16.9mL
(b) (2 points)
$c_{stock}\ V_{stock} = C_{dilute}\ V_{dilute}$　　C_{dilute} = 10.0/1000*14.8/0.500 = 0.296 M
(c) (3 points)
$Sr(OH)_2\ (aq) + 2\ HNO_3\ (aq) \rightarrow Sr(NO_3)_2\ (aq) + 2\ H_2O\ (l)$
(d) (3 points)
$[Sr(OH)_2(aq)]$ = 12.50/(87.62+2*1.00794+2*15.9994)/(50.00/1000) = 2.055 M
1 mole $Sr(OH)_2(aq)$ reacts with 2 mole $HNO_3(aq)$
$c[HNO_3(aq)]$ = 2 *23.90 * 2.055/37.50 = 2.619 M

Concentration (molarity) of the acid is 2.619 M.

C2)

(a) (2 points)

λ = ch / E = 3.00 x 10^{8} x (6.94 × 10^{-19}) / (6.62 x 10^{-34}) = 2.86 × 10^{-7} m OR 286 nm

(b) (3 points)

No, it isn't possible. Visible light has longer wavelength and doesn't have enough energy to emit electron from titanium metal.

(c) (2 points)

E_{263} = hc/1= 6.62 × 10^{34} × 3.00 × 10^{8} - (263 × 10^{-9} m) = 7.55 x 10^{-19} J

E_{k} = E_{263} - E = 7.55 x 10^{-19} J - 6.94 x 10^{-19} J = 6.1 × 10^{-20} J/electron

(d) (2 points)

N = E_{total} / E = 2.30 x 10^{-6} J / (6.94 × 10^{-19} J) = 3.31 × 10^{12}

C3)

(a) (2 points)

$q_1 = m_1c_1\Delta T_1$ = 121.0*0.385*(30.1-100.4) = -3274.9 J = 3.27*10^{3} J

The amount of heat lost by the copper block is 3.27*10^{3} J

(b) (2 points)

$q_2 = m_2c_2\Delta T_2$ = 150.0*4.18*(30.1-25.1)= 3135 J = 3.14 * 10^{3} J

The amount of heat gained by the water is 3.14 * 10^{3} J

(c) (3 points) C= q_3 / ΔT_2 = (3274.9-3135)/(30.1-25.1)=27.98 J/K = 28 J/K

The heat capacity of the calorimeter is 28 J/K

(d) (2 points)　 $-q_1 = q_2$

121.0*0.385*(T_f - 100.4) = 150.0*4. 18* (T_f - 25.1)

T_f = 30.307 ° C = 30.3 ° C　　　The final temperature of the system 30.3 ° C.

C4)

(a) (4 points) First ionization energy (I_1) of Ne: Ne(g) -> Ne (g) + e $\Delta H>0$

[He]$2s^22p^5$　[He]$2s^22p^6$

Electronic affinity (E_1) of F: F(g) + e -> F^-(g)　$\Delta H<0$

[He]$2s^22p^5$　　[He]$2s^22p^6$

(b) (2 points) I_1 [Ne] will be positive, and E_1 [F] will be negative.

(c) (3 points) No. I_1 [Ne] would be larger.

The Z (and Z_{eff}) for Ne is greater than the Z (and Z_{eff}) for F, so the nucleus of Ne has greater attractive force to its valence electron. And I_1 [Ne] is greater than E_1 [F].